

Using ARM CPU Architecture in Production

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What is ARM ?

- Advanced RISC Machine
 - x86: CISC (Complex Instruction Set Computers)
 - ARM: RISC (Reduced Instruction Set Computers)

Multiplying Two Numbers in Memory

1	2	3	4
2			
3			
4			
5			
6			

Memory

A	D
B	E
C	F

Register

MULT 2:3, 5:2

CISC

LOAD A, 2:3

LOAD B, 5:2

PROD A, B

STORE 2:3, A

RISC

CISC vs. RISC

CISC	RISC
Emphasis on hardware	Emphasis on software
Includes multi-clock complex instructions	Single-clock, reduced instruction only
Memory-to-memory: "LOAD" and "STORE" incorporated in instructions	Register to register: "LOAD" and "STORE" are independent instructions
Small code sizes, high cycles per second	Low cycles per second, large code sizes
Transistors used for storing complex instructions	Spends more transistors on memory registers

RISC Advantages

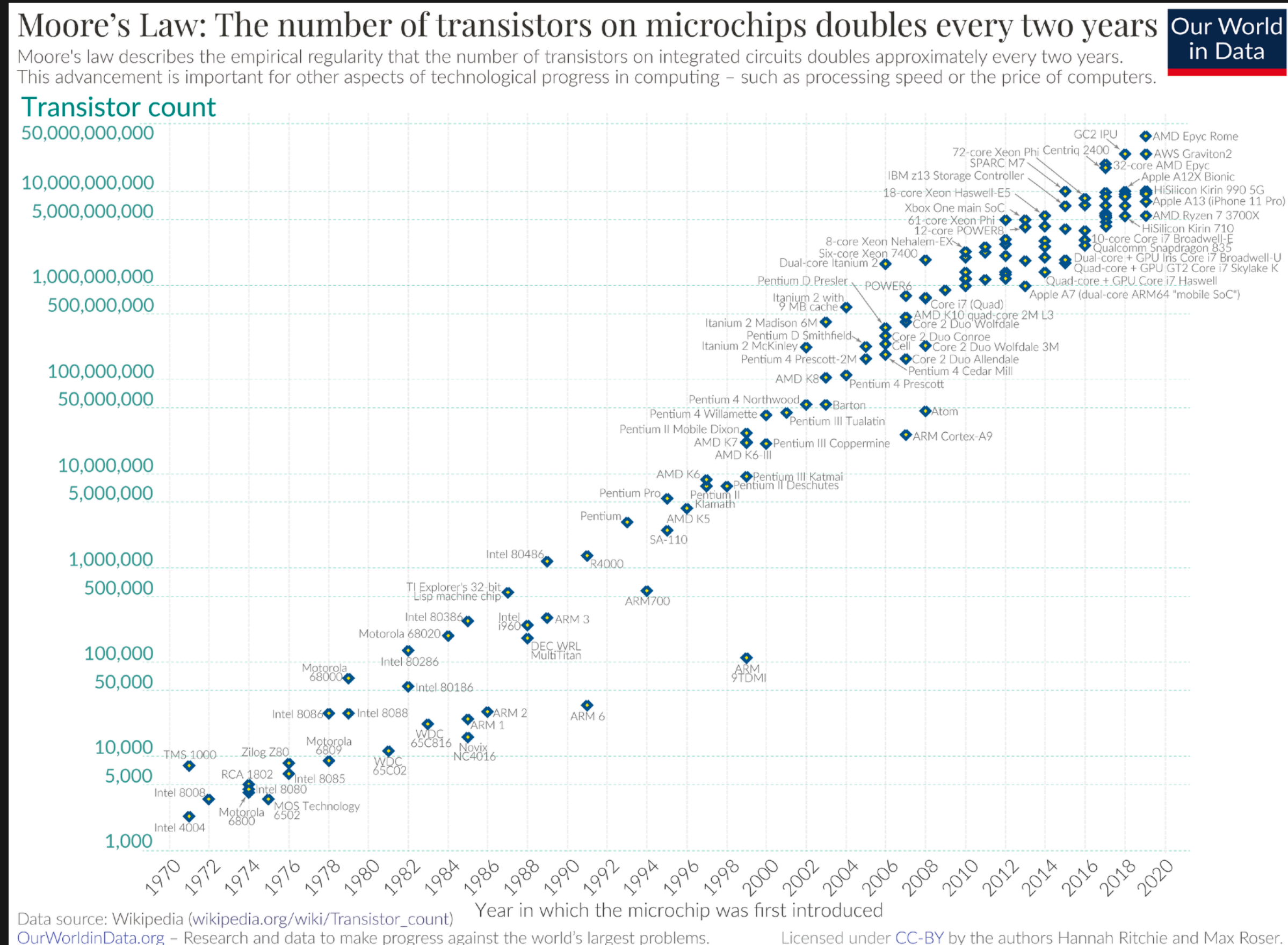
- More lines of code, less efficient ?
- Pipelining: one clock cycle each command
- Less transistors: leaving more room for general purpose registers
- Less re-load: the operand will remain in the register until another value is loaded in its place
 - CISC: the processor automatically erases the registers after “MULT” executed

Marketing

- x86 is more common in the server market
- ARM architectures are used in mobile devices and laptops
 - Lower power consumption
 - Lower heat generation

Moore's Law

- “The number of transistors in an integrated circuit (IC) doubles about every two years.”
- 5nm -> 3nm -> ...



AWS Graviton (ARM)

- Another racetrack
- 7nm - 30 billion transistors
- Large 64KB L1 cache and 1MB L2 Cache per vCPU
 - Intel i7 6700: 32KB L1 / 256KB L2
- Every vCPU is a physical core, no Hyper-Threading
- Etc...

AWS Graviton Advantages



- Improved performance (+20%)
- Lower instance costs (-20%)
- Energy efficiency
- Cost performance ratio

AWS Graviton Advantages

- Billing 2023-07
- EC2 / ElasticCache / ElasticSearch / Staging / RDS
- Unit: CNY
- Optimal estimaition

Env	Cost	Saving
Sandbox	88,975.75	
Staging	18,666.42	
Meican1	671,440.96	
Meican2	110,939.22	
Total	890,022.35	178,004.47

Benchmarking

Server 1 vCPU, Client 100 connections

Arch	Total Requests	RPS	Avg. Latency	p50 Latency	p99 Latency
amd64	1761056	14.75k	15.70ms	1.15ms	71.54ms
arm64	2018862 (+14.6%)	16.89k (+14.5%)	12.92ms (-17.7%)	1.22ms (+5.7%)	63.55ms (-12.5%)

Server 2 vCPU, Client 200 connections

Arch	Total Requests	RPS	Avg. Latency	p50 Latency	p99 Latency
amd64	3521897	29.48k	8.06ms	1.51ms	45.26ms
arm64	4245043 (+20.5%)	35.54k (+20.5%)	6.04ms (-25.1%)	1.09ms (-38.5%)	37.42ms (-20.9%)

Go WebServer (High Load)

Benchmarking

Server 2 vCPU, Client 10 connections

Arch	Total Requests	RPS	Avg. Latency	p50 Latency	p99 Latency
amd64	2772679	23.19k	562.29us	179.00us	9.44ms
arm64	3059087 (+10.3%)	25.62k (+10.4%)	205.77us (-63.4%)	174.00us (-2.8%)	803.00us (-91.5%)

Go WebServer (Low Load)

Benchmarking

- MySQL by @maliang: <https://planetmeican.atlassian.net/browse/OOPS-1423>
- Sysbench by Percona: <https://www.percona.com/blog/comparing-graviton-performance-to-arm-and-intel-for-mysql/>

Benchmarking

- ARM CPUs show better results on EC2 with more vCPU and with higher load, especially in high-concurrency scenarios
- Physical cores: no throttling
- Perfect for Web scenario

Pitfalls

- Graviton 2 RSA performance issue (@maliang)
 - Sign: ok
 - Check: -50% performance
 - <https://planetmeican.atlassian.net/browse/OOPS-1423>
- Graviton 3: additional core to resolve the issue

Building ARM Container Image



- Requirement:
 - Go $\geq$ 1.9, support for Linux ARM 64bit (<https://go.dev/blog/ports>)

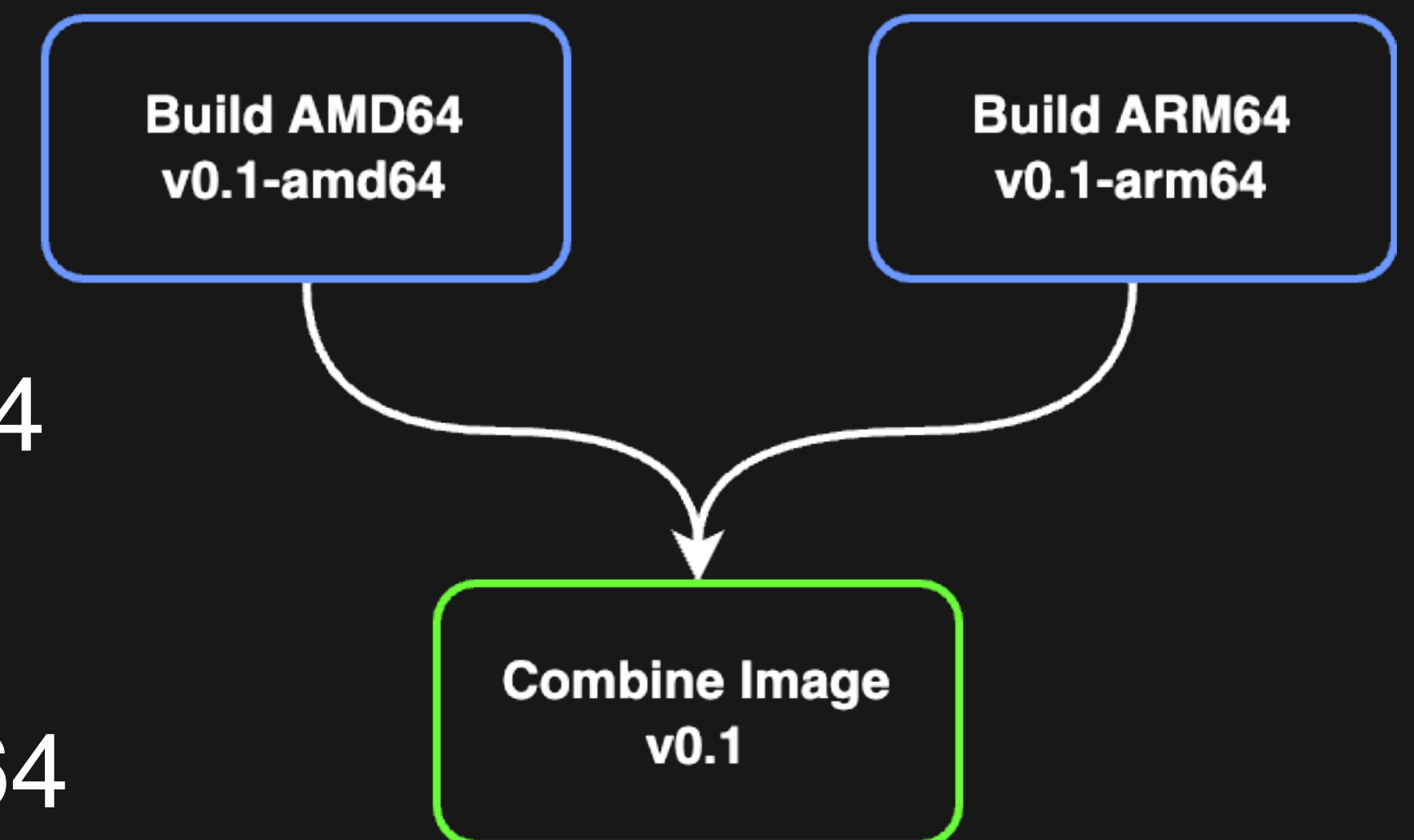
Building ARM Container Image

- Local environment
- `docker buildx build --platform linux/arm64,linux/amd64 -t misc:test -f Dockerfile .`
- nerdctl (hard to use on MacOS)

Building ARM Container Image



- GitlabCI: prefer not to use dind
- Jobs:
 - compile_upload_ecr_sandbox_master_arm64
 - On arm64 machine (tags: go-arm64)
 - compile_upload_ecr_sandbox_master_amd64
 - On amd64 machine (tags: go)
 - combine_image_for_branch





Building ARM Container Image

```
compile_upload_ecr_sandbox_main_amd64:
  stage: build_release
  image:
    name: 651844176281.dkr.ecr.cn-northwest-1.amazonaws.com.cn/gcr.io/kaniko-project/executor:v1.9.2-debug
    entrypoint: [""]
  script:
    - RELEASE_VERSION=v$(cat next-version)
    - echo $RELEASE_VERSION
    - GIT_VERSION=$CI_COMMIT_SHORT_SHA
    - BUILD_TIME=$(date '+%Y-%m-%d %H:%M:%S')
    - BUILD_HOST=$(hostname)
    - echo "{\"credsStore\":\"ecr-login\",\"credHelpers\":{\"$SANDBOX_ECR_PATH\":\"ecr-login\"}}\" > /kaniko/.
    - /kaniko/executor
    - --context $CI_PROJECT_DIR
    - --dockerfile $CI_PROJECT_DIR/docker/Dockerfile_sandbox
    - --build-arg GIT_VERSION=$GIT_VERSION
    - --build-arg GIT_TAG=$RELEASE_VERSION
    - --build-arg BUILD_TIME="$BUILD_TIME"
    - --build-arg BUILD_HOST=$BUILD_HOST
    - --destination $SANDBOX_ECR_PATH/$PROJECT_NAME:$RELEASE_VERSION-amd64
  when: manual
  allow_failure: false
  only:
    refs:
      - tags
  tags:
    - go
    - sandbox
```

```
compile_upload_ecr_sandbox_main_arm64:
  stage: build_release
  image:
    name: 651844176281.dkr.ecr.cn-northwest-1.amazonaws.com.cn/gcr.io/kaniko-project/ex
    entrypoint: [""]
  script:
    - RELEASE_VERSION=v$(cat next-version)
    - echo $RELEASE_VERSION
    - GIT_VERSION=$CI_COMMIT_SHORT_SHA
    - BUILD_TIME=$(date '+%Y-%m-%d %H:%M:%S')
    - BUILD_HOST=$(hostname)
    - echo "{\"credsStore\":\"ecr-login\",\"credHelpers\":{\"$SANDBOX_ECR_PATH\":\"ecr-
    - /kaniko/executor
    - --context $CI_PROJECT_DIR
    - --dockerfile $CI_PROJECT_DIR/docker/Dockerfile_sandbox
    - --build-arg GIT_VERSION=$GIT_VERSION
    - --build-arg GIT_TAG=$RELEASE_VERSION
    - --build-arg BUILD_TIME="$BUILD_TIME"
    - --build-arg BUILD_HOST=$BUILD_HOST
    - --destination $SANDBOX_ECR_PATH/$PROJECT_NAME:$RELEASE_VERSION-arm64
  when: manual
  allow_failure: false
  only:
    refs:
      - tags
  tags:
    - go-arm64
    - sandbox
```

Building ARM Container Image

```
combine_image_for_sandbox_main:
  stage: combine_image_for_release
  image:
    name: 651844176281.dkr.ecr.cn-northwest-1.amazonaws.com.cn/titan/base-images/manifest-tool:alpine-v2.0.8
    entrypoint: ["/bin/sh"]
  script:
    - RELEASE_VERSION=v$(cat next-version)
    - echo $RELEASE_VERSION
    - echo "{\"credsStore\":\"ecr-login\",\"credHelpers\":{\"$SANDBOX_ECR_PATH\":\"ecr-login\"}}\" > /.docker/config.json
    - manifest-tool --docker-cfg /.docker/config.json push from-args
    - --platforms linux/amd64,linux/arm64
    - --template $SANDBOX_ECR_PATH/$PROJECT_NAME:$RELEASE_VERSION-ARCH
    - --tags $RELEASE_VERSION
    - --target $SANDBOX_ECR_PATH/$PROJECT_NAME:$RELEASE_VERSION
  dependencies:
    - init
    - compile_upload_ecr_sandbox_main_amd64
    - compile_upload_ecr_sandbox_main_arm64
  needs:
    - init
    - compile_upload_ecr_sandbox_main_amd64
    - compile_upload_ecr_sandbox_main_arm64
  only:
  refs:
    - tags
  tags:
    - go-arm64
    - sandbox
```

ARCH is a template variable
stands for arm64 / amd64

Auto-triggered when image builds finished

Using GitlabCI “include”

- Predefined jobs
- 600+ lines -> 20+ lines
- Kept updated

```
variables:
  PROJECT_NAME: titan/alertmanager-bridge
  REPO_NAME: go.planetmeican.com/titan/alertmanager-bridge

include:
  - project: 'meican-cd/ci-template'
    ref: main
    file: all-in-one-go1.20.yml

stages:
  - init
  - lint
  - analysis
  - report
  - build_branch
  - combine_image_for_branch
  - release
  - build_release
  - combine_image_for_release
```

Parameterized Templates - Variables

```
variables:
  PROJECT_NAME: titan/pagerduty
  REPO_NAME: go.planetmeican.com/titan/pagerduty
  ANALYSIS_TIMEOUT: 1m
  ANALYSIS_GOFLAGS: -mod=vendor
```

.gitlab-ci.yml

```
# Optional variables
# -- TEST_TIMEOUT: for example 1m
# -- TEST_GOFLAGS: for example, -mod=vendor
variables:
  ANALYSIS_TIMEOUT: 1m
  ANALYSIS_GOFLAGS: -mod=vendor

test:go_test:
  image: 651844176281.dkr.ecr.cn-northwest-1.amazonaws.com.cn/titan/base-images/gitlab-ci-go:v1.20.6-alpine3.18-mod-token
  stage: analysis
  script:
    - export GOFLAGS="$ANALYSIS_GOFLAGS"
    - CGO_ENABLED=0 go test -p 1 -v -test.short -coverprofile=coverage.out -timeout="$ANALYSIS_TIMEOUT" $(go list ./... | grep -
  artifacts:
    when: always
    expire_in: 1 day
    paths:
      - report.json
      - coverage.out
      - checkstyle.xml
  except:
    - tags
  tags:
    - go-arm64
    - sandbox
```

jobs/analysis-go1.20-test.yml



Parameterized Templates - Override

```
# override the template
test:go_test:
  services:
    - 651844176281.dkr.ecr.cn-northwest-1.amazonaws.com.cn/titan/base-images/gitlab-ci-go:v1.20.6-alpine3.18-mod-token
    - 651844176281.dkr.ecr.cn-northwest-1.amazonaws.com.cn/redis:6.2
```

.gitlab-ci.yml

```
# Optional variables
# -- TEST_TIMEOUT: for example 1m
# -- TEST_GOFLAGS: for example, -mod=vendor
variables:
  ANALYSIS_TIMEOUT: 1m
  ANALYSIS_GOFLAGS: -mod=vendor

test:go_test:
  image: 651844176281.dkr.ecr.cn-northwest-1.amazonaws.com.cn/titan/base-images/gitlab-ci-go:v1.20.6-alpine3.18-mod-token
  stage: analysis
  script:
    - export GOFLAGS="$ANALYSIS_GOFLAGS"
    - CGO_ENABLED=0 go test -p 1 -v -test.short -coverprofile=coverage.out -timeout="$ANALYSIS_TIMEOUT" $(go list ./... | grep -
```

```
ways
  artifacts:
    paths:
      - coverage.out
      - style.xml
    expire_in: 1 day
  tags:
    - go-arm64
    - sandbox
```

jobs/analysis-go1.20-test.yml

Gitlab CI Template

- <https://gitlab.planetmeican.com/meican-cd/ci-template>

Building ARM Container Image

```
FROM 651844176281.dkr.ecr.cn-northwest-1.amazonaws.com.cn/titan/base-images/alpine-tz8:v3.18.2

COPY --from=build /executor /executor
COPY --from=build /go/src/public /public

RUN addgroup go \
    && adduser -D -G go go \
    && chown -R go:go /executor

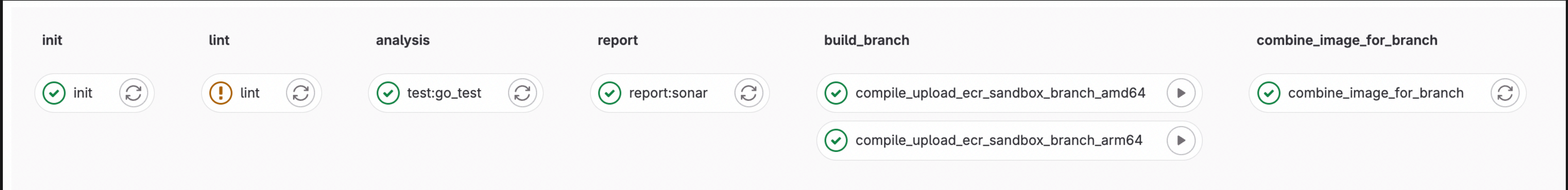
# metrics port
EXPOSE 18888
EXPOSE 8023
EXPOSE 8024

ENTRYPOINT ["/executor"]
```

Building ARM Container Image

- Make sure the base image support multi-arch
 - “docker manifest inspect”
- Run job in specific arch:
 - “tag: go-arm64”
- Use predefined CI templates

```
{
  "schemaVersion": 2,
  "mediaType": "application/vnd.docker.distribution.manifest.list.v2+json",
  "manifests": [
    {
      "mediaType": "application/vnd.docker.distribution.manifest.v2+json",
      "size": 1571,
      "digest": "sha256:8961ca607db2a0c80010aeb5fc9bed0253448bc68711eaa14253a392f6c48280",
      "platform": {
        "architecture": "amd64",
        "os": "linux"
      }
    },
    {
      "mediaType": "application/vnd.docker.distribution.manifest.v2+json",
      "size": 1571,
      "digest": "sha256:aebc052b5a7f50d79f00a4deee22109e2ecf22f439140a03f2c9770055ffec7b",
      "platform": {
        "architecture": "arm64",
        "os": "linux",
        "variant": "v8"
      }
    }
  ]
}
```

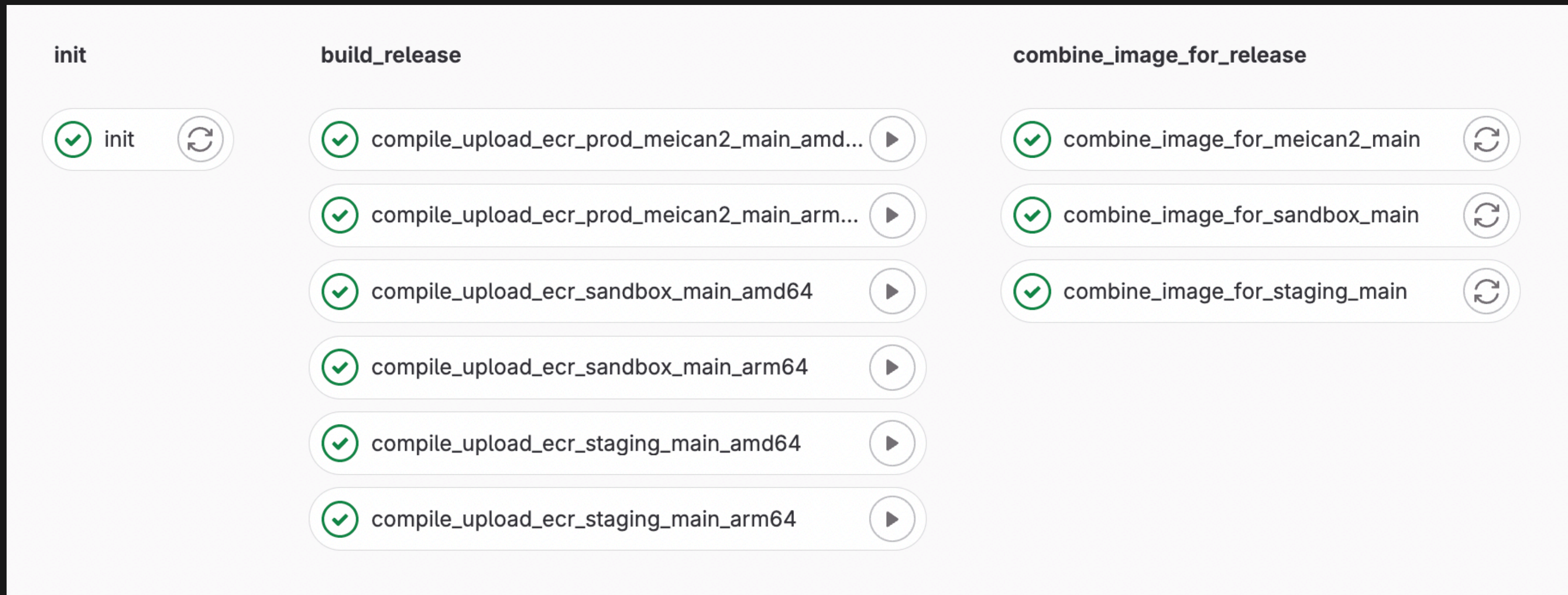


Manually triggered

Auto

GitlabCI

Workflow



Manually triggered

Auto

GitlabCI

Workflow

Deployment

- nodeSelector:
 - kubernetes.io/arch (optional)
- tolerances:
 - Allow scheduled on ARM64 instances
 - Taint on ARM64 instances

```
nodeSelector:
  intend: app1
  kubernetes.io/arch: arm64
tolerations:
  - key: kubernetes.io/arch
    value: arm64
    operator: Equal
    effect: NoExecute
```

Optional

Base Images w/ Multi-Arch Support



- Go lint:
 - titan/base-images/golangci-lint:v1.52.2-alpine-mod-token
- Kaniko:
 - gcr.io/kaniko-project/executor:v1.9.2-debug
- Combine image:
 - titan/base-images/manifest-tool:alpine-v2.0.8
- <https://gitlab.planetmeican.com/titan/base-images>

2023-Jul-21: multi arch support

651844176281.dkr.ecr.cn-northwest-1.amazonaws.com.cn/titan/base-images/alpine-

017039971561.dkr.ecr.cn-northwest-1.amazonaws.com.cn/titan/base-images/alpine-

321659100662.dkr.ecr.cn-northwest-1.amazonaws.com.cn/titan/base-images/alpine-

693295102651.dkr.ecr.cn-northwest-1.amazonaws.com.cn/titan/base-images/alpine-

RELEASE.md / README.md

Important Notes

- In response to FinOps
 - Reduce cloud cost
 - Establish spend model
 - <https://www.oreilly.com/library/view/cloud-finops/9781492054610/ch01.html>
- Migrate to ARM64:
 - Final warning on 2023-12-01
 - Deadline: 2023-12-31

References

- <https://planetmeican.atlassian.net/wiki/spaces/DEV/pages/620462082/ARM+CPU>

Fin.
Question ?